

Auteur: Siebe Mees
Studierichting: Informatica
Oefeningenreeks 4A nr. 31.5

$$\int (x+3)\sqrt[3]{x+2}dx$$

$$t = x + 2 (\Rightarrow dt = dx \Rightarrow x = t - 2)$$

$$\int (t-2+3)t^{\frac{1}{3}}dt$$

$$= \int (t+1)t^{\frac{1}{3}}dt$$

$$= \int \left(t^{\frac{4}{3}} + t^{\frac{1}{3}} \right) dt$$

$$= \frac{t^{\frac{7}{3}}}{\frac{7}{3}} + \frac{t^{\frac{4}{3}}}{\frac{4}{3}} + C$$

$$= \frac{3}{7}t^{\frac{7}{3}} + \frac{3}{4}t^{\frac{4}{3}} + C$$

$$x+2=t$$

$$= \frac{3}{7}(x+2)^{\frac{7}{3}} + \frac{3}{4}(x+2)^{\frac{4}{3}} + C$$

References